

ENTERPRISE WORKFORCE MANAGEMENT EDITION

Human Resources & Workforce Management

Complete Operational Architecture, Workflows, Biometrics, Payroll & Lifecycle Guide

An authoritative, end-to-end standard operating manual covering the complete HR ecosystem across all four operating companies: Multi-Company Tenancy, Organization & Department Hierarchy, Employee Master Profiles, Versioned Document Vault, Attendance & Shift Scheduling, ZKTeco Biometric Push/Pull Ingestion, Double-Entry Leave Ledger, 2-Tier Department Operations & Daily Work Reporting, Employee Financing Subledger, Attendance-Driven Payroll Processing, Fixed Asset Custody, Performance Appraisals, Separation & Clearance, and Source-Reconciled Final Settlements.

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Multi-Company Operating Group Scope:

- BMC Construction
- YMC Construction
- 7 Orbit
- 7 Orbit Medical Billing

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1. Complete HR Architecture & Organizational Hierarchy

The **Human Resources & Workforce Management Module** of the YM Construction Management System (YM-CMS) provides a multi-company, lifecycle-driven human capital architecture. It is specifically designed to meet the rigorous compliance, organizational governance, cost-allocation, and biometric tracking requirements of construction contracting, IT technology ventures, medical billing operations, and group corporate administration.

1.1 Multi-Tenant Corporate Structure & Tenancy Isolation

YM-CMS operates four distinct, legally independent operating companies within a single unified Laravel and Filament framework. None of these entities is configured as a synthetic tenant or database-level subsidiary; each maintains completely isolated memberships, organizational structures, chart of accounts, employee master records, attendance logs, and financial ledgers:

- **BMC Construction:** Civil engineering, infrastructure development, heavy construction contracting, and site-based project operations.
- **YMC Construction:** General contracting, architectural development, project subcontracting, and construction procurement.
- **7 Orbit:** Digital marketing, software engineering, media production, creative branding, and technology solutions.
- **7 Orbit Medical Billing:** Revenue cycle management (RCM), medical coding, healthcare billing claims, and professional services.

Tenancy & Security Invariant

All HR resources enforce strict tenant scoping via `company_id`. Cross-company visibility, document sharing, or reporting lines are prohibited at both the database foreign-key layer and Eloquent policy boundaries. Group-level reporting is permitted solely through authorized multi-company aggregation interfaces for users possessing explicit group-wide authorization.

1.2 Separation of Global Employee Identity vs Company-Specific Employment

A foundational architectural innovation in YM-CMS is the strict separation between an individual's **Global Personal Identity** (`Employee`) and their **Company-Specific Operational Contract** (`Employment`):

```
+-----+
| GLOBAL EMPLOYEE IDENTITY (app/Models/Employee.php) |
| - Full Legal Name, Date of Birth, Gender, Marital Status, Blood Group |
| - Normalized CNIC (Encrypted at Rest with SHA-256 Keyed Lookup Hash) |
| - Profile Picture, Global Contact Info, Emergency Contacts, Qualifications, Past Exp |
+-----+
      |
      +-----+
      | (1-to-Many Multi-Company Employments) |
      +-----+
      v               v
+-----+ +-----+
| BMC EMPLOYMENT (Employment.php) | | 7 ORBIT EMPLOYMENT (Employment.php) |
| - Company: BMC Construction      | | - Company: 7 Orbit                  |
| - Code: EMP-00001                | | - Code: EMP-00001 (Isolated)         |
| - Dept: Civil Engineering         | | - Dept: Web Development            |
| - Designation: Project Manager   | | - Designation: Senior Tech Lead    |
| - Category: Project Staff        | | - Category: Administrative Staff    |
| - Type: Permanent                | | - Type: Contract                   |
| - Status: Active                 | | - Status: Active                    |
| - Compensation: PKR 150,000 / mo | | - Compensation: PKR 180,000 / mo      |
| - Shift: Construction Site Shift | | - Shift: Head Office Tech Shift    |
+-----+ +-----+
```

This dual-entity design enables a single person (e.g., an executive engineer or IT director) to hold simultaneous or sequential employments across different sister companies without duplicating their personal profile, while guaranteeing that payroll, attendance, loans, taxes, and departmental reporting remain 100% isolated within each respective legal company.

1.3 Department Hierarchy & Nested Parent-Child Architecture

Departments in YM-CMS (`Department`) are company-scoped organizational units supporting unlimited recursive parent-child hierarchy. A department can be a root-level division (e.g., *Operations*, *Finance & Accounts*) or a sub-department (e.g., *Site Logistics* under *Operations*).

- **Self-Parenting Prevention:** A department cannot be assigned as its own parent.
- **Cycle Prevention:** Recursive ancestor scanning blocks circular dependencies (e.g., A $\rightarrow$ B $\rightarrow$ C $\rightarrow$ A).
- **Same-Company Enforcement:** Parent departments must strictly belong to the same active company tenant.
- **Child-Aware Deletion Constraint:** A department cannot be deleted if it has child departments or active employments assigned to it.

1.4 Designations & Departmental Linkages

Designations (`Designation`) represent formal corporate job titles (e.g., *Chief Operating Officer*, *Site Engineer*, *Full Stack Developer*, *Medical Billing Specialist*). Designations are company-scoped and can optionally be linked to a specific department. In the Employment creation interface, selecting a Department dynamically filters the available Designations to ensure organizational consistency.

1.5 Work Locations & Project Site Linkages

Work Locations (`WorkLocation`) establish controlled physical or virtual job sites for staff assignment. Each location captures a location code, location name, physical address, and an optional link to an active same-company **Project Site** (`ProjectSite`). This allows attendance devices, site allowances, and project labor cost distributions to automatically bind to specific physical work locations.

1.6 Department Teams & Specialized Pods

Under Phase D additions, Departments can be further structured into operational **Department Teams** (`DepartmentTeam`). Teams allow fine-grained task delegation, daily check-in supervision, and specialized creative/technical workflows. Supported team types include:

TEAM TYPE	DOMAIN FOCUS	SPECIALIZED TRACKING SCHEMA & FEATURES
Social Media	Digital Marketing & Growth	8-stage content pipeline, platform targeting (Meta, LinkedIn, X, TikTok), caption bank, scheduling.
Graphic Design	Visual Creative & Branding	Design types (Post, Banner, Brochure, UI), dimensions/aspect ratios, brand briefs, Figma/PSD deliverables.
Video Production	Reels, Promos & Walkthroughs	Video formats, target durations, cloud footage URLs, script notes, Draft V1/V2 & Final Cut links.
Web Development	Software Engineering	Task categories, GitHub PR links, staging/live URLs, QA testing statuses, bug tracking.
Sales & CRM	Outreach & Client Acquisition	Daily outreach KPIs (leads received, calls made, meetings booked, proposals sent, closed revenue).
General / Operations	Standard Department Work	Standard task assignment, deliverable attachments, deadline tracking, and blocker escalations.

1.7 Employee Categorization & Employment Types

To satisfy accounting, payroll distribution, and statutory reporting rules, every Employment is categorized into two distinct orthogonal classifications:

CLASSIFICATION AXIS	ALLOWED ENUM VALUES	OPERATIONAL & ACCOUNTING SIGNIFICANCE
Employment Category <code>EmploymentCategory</code>	• <code>DIRECTOR</code> • <code>ADMINISTRATIVE STAFF</code> • <code>PROJECT STAFF</code>	Critical Accounting Split: Administrative Staff and Directors post salary expenses directly to Head Office overhead accounts (<code>5010 Salaries Expense</code>). Project Staff require mandatory project and cost-center allocations, posting directly to <code>7000 Construction Direct Labor Cost</code> .

CLASSIFICATION AXIS	ALLOWED ENUM VALUES	OPERATIONAL & ACCOUNTING SIGNIFICANCE
Employment Type EmploymentType	- PERMANENT - CONTRACT - DAILY WAGES - INTERNSHIP	Governs leave accrual eligibility, probation requirements, notice period mandates, and benefits entitlement in final settlement calculations.
Employment Status EmploymentStatus	- PROBATION - ACTIVE - ON LEAVE - RESIGNED - TERMINATED - ENDED (Legacy Read-Only)	Determines active presence in biometric sync, daily attendance processing, monthly payroll generation, task delegation eligibility, and clearance readiness.

1.8 Atomic Sequential Employee Code Allocation Engine

Employee Codes (e.g., EMP-00001) are generated dynamically and atomically per company using the EmployeeCodeSequence model and AllocateEmployeeCodeAction:

- **Row Locking:** The company's sequence record is locked using database-level pessimistic locking (lockForUpdate()) within an active transaction.
- **Collision Avoidance:** If a manually entered or legacy code collides with the sequence, the engine scans forward and skips to the next available unique code.
- **Configurable Formatting:** The prefix (default EMP) and zero-padding width (default 5 digits) can be customized per company tenant via the *Employee Code Sequences* configuration resource.
- **Tenancy Isolation:** Sequence counters are strictly isolated; Company A and Company B can both maintain an independent EMP-00001 without conflict.

1.9 Comprehensive HR Entity-Relationship Architecture Diagram



2. Employee Master & Complete Profile Management

The Employee Master is the central repository of personnel data within YM-CMS. It provides HR administrators, department heads, and compliance officers with complete 360-degree visibility over an individual's personal identity, emergency contacts, academic qualifications, employment history, bank accounts, approved salary structure, asset custodies, performance records, and legal document compliance.

2.1 Employee Master Architecture & Data Flow

When creating a new worker, the system automatically performs an atomic transaction that creates the global `Employee` record and instantiates their primary `Employment` within the active company tenant.

Software Navigation: Creating a New Employee Record

1

Sidebar Navigation: Expand HR Management and select Employees (`/admin/company/{tenant}/employees`).

2

Action: Click the primary + New Employee button in the header toolbar.

3

Tab 1: Personal Information: Enter First Name, Last Name, Father/Husband Name, CNIC, Date of Birth, Gender, Marital Status, Blood Group, Personal Email, Mobile Number, and Residential Address. Upload a profile photograph (JPEG/PNG, max 2MB).

4

Tab 2: Employment Details: Select Department, Designation, Employment Category (Director, Administrative Staff, Project Staff), Employment Type (Permanent, Contract, Daily Wages, Internship), Joining Date, Reporting Manager, Work Location, and Work Schedule. Set Employee Code to *Auto-generated* or input a manual override.

5

Save & Commit: Click Create. The system transactionally commits the Employee, generates the atomic Employee Code, links the initial Employment, logs the creation audit event, and redirects to the full Profile View.

2.2 Personal Identity, PII Encryption & Keyed Hashes

To comply with strict enterprise data privacy standards, all Personally Identifiable Information (PII) is protected using envelope encryption at rest:

Field Identifier	Database Column & Cast	Encryption, Normalization & Security Controls
National ID (CNIC)	<code>cnic</code> (encrypted:string) <code>cnic_hash</code> (indexed string)	- Automatically stripped of dashes (normalized to 13 digits: <code>3520112345671</code>). - Encrypted at rest via Laravel AES-256-CBC cipher. - Supported by a keyed HMAC-SHA256 hash (<code>cnic_hash</code>) for instant duplicate detection and search lookups without plaintext exposure. - Masked on screens (<code>35201-****71-1</code>) unless the user possesses the <code>ViewSensitiveData:Employee</code> permission.
Bank Account & IBAN	<code>account_number</code> (encrypted) <code>iban</code> (encrypted)	- Fully encrypted at rest in <code>employee_bank_accounts</code> . - Masked in general UI views (<code>PK** **** ** 1234</code>). - Excluded from standard Spatie activity log payload snapshots to prevent log leakage.
Private Health & Medical	<code>medical_notes</code> (encrypted)	- Accessible only to users holding the specialized <code>ViewMedicalInfo:Employee</code> permission. - Hidden from department heads and team leads.

2.3 Company Employment Attributes & Lifecycle Dates

An individual's active Employment record governs their operational lifecycle. Important fields include:

- **Joining Date (`joining_date`)**: The official start date. Used to prorate basic salary in the first payroll month and determine leave accrual eligibility.
- **Probation Start & End (`probation_start_date` , `probation_end_date`)**: Explicit duration of trial employment. When probation is successfully completed, HR records the **Confirmation Date** (`confirmation_date`), moving status from `Probation` to `Active` .
- **Notice Period Days (`notice_period_days`)**: Calendar days of advance notice required upon resignation. Used during separation and final settlement notice-pay shortfall calculations.
- **Reporting Manager (`reporting_to_employment_id`)**: Points to another active Employment within the same company. The system automatically prevents self-reporting and circular management loops.
- **Employment Change History (`employment_changes`)**: Any modification to an employee's department, designation, manager, work location, or status automatically creates an immutable, append-only history record capturing the old value, new value, effective date, and acting user ID.

2.4 Profile Infolists & Tabbed Sub-Masters

The Employee Profile infolist provides dedicated relation managers for managing detailed personnel records:

SUB-MASTER TAB	MANAGED FIELDS	BUSINESS RULES & VALIDATION CONSTRAINTS
Emergency Contacts (<code>EmployeeEmergencyContact</code>)	Contact Name, Relationship, Primary Phone, Secondary Phone, Work Phone, Address, Is Primary Flag.	Exactly one primary emergency contact is enforced per employee. Secondary phone and full address are masked without sensitive data authorization.
Qualifications (<code>EmployeeQualification</code>)	Degree/Diploma Title, Major/Field of Study, Institution Name, Passing Year, Total Marks/CGPA, Obtained Marks, Grade.	Passing year cannot be in the future. Multiple degrees are ordered chronologically. Used for HR verification and compliance audits.
Previous Experience (<code>EmployeeExperience</code>)	Organization Name, Designation Held, Starting Date, Ending Date, Reason for Leaving, Contact Reference.	Ending date cannot precede starting date. Total prior service years are computed automatically for seniority calculations.
Bank Accounts (<code>EmployeeBankAccount</code>)	Bank Name, Branch Name/Code, Account Title, Account Number, IBAN, Is Primary Payroll Flag.	Enforces exactly one active primary payroll bank account per employment. Account number and IBAN are encrypted at rest. Used by Payroll Bank Advice exports.
Compensation History (<code>EmploymentCompensation</code>)	Monthly Basic Salary, House & Travel, Food, Fuel, Mobile, Internet, Site, Project, Other Allowances, Gross Salary.	Effective-dated salary records. Approved salaries are 100% immutable. Approving a new compensation automatically closes the previous period.
Document Vault (<code>Document</code>)	Controlled Document Type, Title, Issue Date, Expiry Date, Verification Status, Version History, File Attachment.	Private local storage. Uploading replacement files increments version counter (<code>v1</code> ? <code>v2</code>) without overwriting historical files.

3. Typed HR Documents & Compliance Management (HR-2)

Phase HR-2 establishes an enterprise-grade document compliance subsystem built directly on top of the YM-CMS private document platform. It standardizes corporate onboarding documentation, enforces cryptographic integrity, supports multi-tier verifications, and highlights missing mandatory compliance records.

3.1 Controlled Document Types & Applicability Scopes

The system standardizes personnel documentation into six controlled, company-configurable document types (`HrDocumentType`):

DOCUMENT TYPE CODE	APPLICABILITY SCOPE	DEFAULT SENSITIVITY	REQUIRED ATTRIBUTES & COMPLIANCE RULES
CNIC	EMPLOYEE (GLOBAL)	RESTRICTED	Requires Issue Date and Expiry Date. Requires Identity Document permission to view/download.
Educational Document	EMPLOYEE (GLOBAL)	CONFIDENTIAL	Degree certificate or official transcript. Verified by HR against submitted qualifications.
Experience Certificate	EMPLOYEE (GLOBAL)	CONFIDENTIAL	Relieving letters and service certificates from previous employers.
Appointment Letter	EMPLOYMENT (COMPANY)	RESTRICTED	Company-specific signed employment contract or formal offer letter.
Medical Certificate	EMPLOYEE (GLOBAL)	RESTRICTED	Pre-employment medical fitness certificate. Requires dedicated Medical Data permission.
Police Verification	EMPLOYEE (GLOBAL)	RESTRICTED	Character certificate or formal police clearance. Mandatory for site security access.

3.2 Private Storage, SHA-256 Checksums & Immutable Versions

All HR documents are stored outside the public web root on the private Laravel `local` disk (`storage/app/documents`). Direct HTTP file access is blocked. Files can only be accessed through short-lived, authenticated preview URLs generated dynamically by the document controller.

- **Cryptographic Fingerprinting:** Every uploaded file is hashed using SHA-256 upon ingestion. The resulting 64-character hexadecimal digest is permanently stored in `document_versions.checksum`.
- **Immutable Versioning:** If an employee submits an updated document (e.g., a renewed CNIC or updated degree), uploading the file creates a new child version record (`version_number = 2`). The historical version (`version_number = 1`) remains permanently preserved in storage and auditable in the system ledger.
- **Automatic State Reset:** Creating a new document version automatically resets prior Verification and Approval states back to `Draft`, requiring HR to re-verify the new file.

3.3 Verification & Approval Workflow

HR document governance enforces an independent two-stage review lifecycle:

Document Upload (Draft) ? HR Officer Verification (`VERIFIED`) ? HR Manager Final Sign-Off (`APPROVED`)

Software Navigation: Verifying and Approving an Employee Document

- 1 **Navigate to Document:** Open the Employee Profile ? switch to the Documents tab (or navigate directly to Documents ? Documents).
- 2 **Inspect Attachment:** Click the Preview action to inspect the document in the secure private viewer.
- 3 **Step 1 (Verification):** Click Verify Document. The system records the acting user ID (`verified_by`) and current timestamp (`verified_at`).
- 4 **Step 2 (Approval):** A separate authorized HR Approver reviews the verified document and clicks Approve Document. The status transitions to **APPROVED**.
- 5 **Rejection Handling:** If the document is illegible or invalid, the reviewer clicks Reject Document and types a mandatory rejection reason. The status becomes **REJECTED** and an alert is flagged on the employee's profile.

3.4 Compliance Audit Summary & Missing Document Tracking

The system includes an automated compliance scanner that evaluates each employee's document vault against the company's active `HrDocumentType` configuration where `is_required = true`. Any missing mandatory document is highlighted in amber/red on the Employee View infolist and compiled into the **HR Operational Readiness** audit report.

4. Work Calendars, Shifts & Biometric Attendance Management (HR-3, HR-4, HR-5)

The Attendance Subsystem provides a multi-layer ingestion and calculation engine capable of processing real-time biometric raw punches from on-premise fingerprint terminals, handling complex day and overnight cross-midnight shifts, applying company-specific late/grace policies, supporting maker-checker manual corrections, and compiling immutable monthly attendance summaries for payroll calculation.

4.1 Work Calendars, Shift Schedules & Overnight Shifts

Attendance calculation is governed by three foundational configuration entities:

- **Work Calendars (`WorkCalendar`)**: Define the standard working week per company (e.g., Monday through Saturday, Sunday Rest Day). Links to **Company Holidays (`CompanyHoliday`)** for public, religious, and gazetted non-working days.
- **Work Shifts (`WorkShift`)**: Define shift timing boundaries:
 - **Standard Day Shift**: e.g., 09:00:00 to 18:00:00. Break: 13:00 to 14:00.
 - **Overnight Shift (`is_overnight` = `true`)**: e.g., 20:00:00 to 05:00:00 (next day). The system automatically binds punches occurring early the following morning (00:00 to 06:00) to the previous day's attendance ledger date.
- **Shift Assignments (`ShiftAssignment`)**: Map individual Employments to a specific Work Shift with effective date ranges (`effective_from` , `effective_to`), allowing seamless shift rotations without rewriting attendance history.

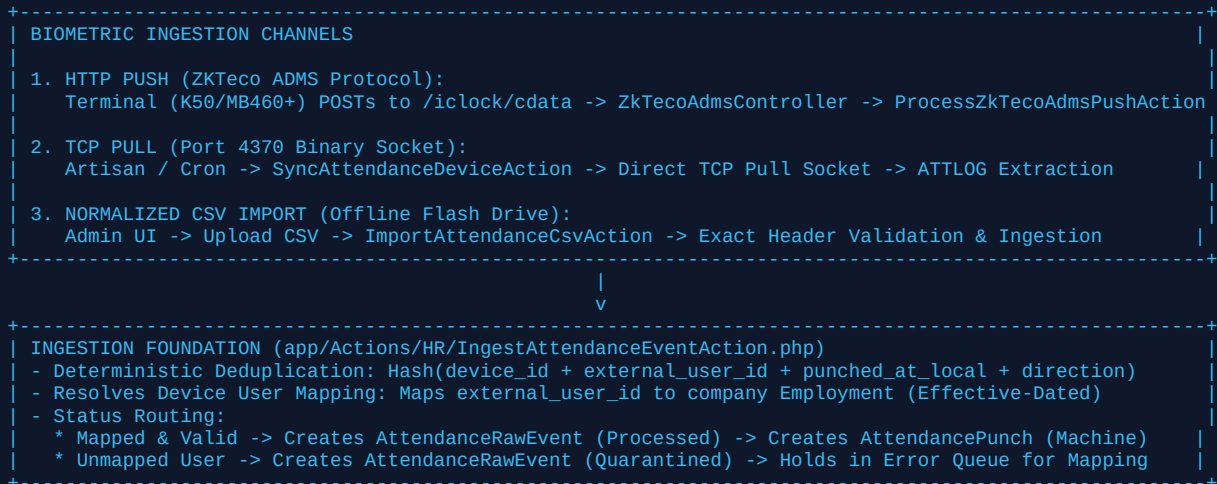
4.2 Effective-Dated Attendance Calculation Rules & Thresholds

Each company configures one or more active **Attendance Rules (`AttendanceRule`)**. The rule defines mathematical criteria for evaluating raw punches into daily attendance states:

RULE PARAMETER	DEFAULT / UNIT	EVALUATION LOGIC & PAYROLL CONSEQUENCE
Grace Period Minutes (<code>grace_minutes</code>)	15 Minutes	Permitted arrival window after shift start. Arrival within shift start + grace minutes is marked PRESENT without penalty.
Late Arrival Threshold (<code>late_threshold_minutes</code>)	16 to 120 Minutes	Arrival after grace period but before half-day threshold. Flags attendance state as LATE and records total late minutes.
Half-Day Minimum Minutes (<code>half_day_minimum_minutes</code>)	240 Minutes (4 Hours)	Total work duration required to earn a half-day credit. Working less than this threshold automatically marks the day as ABSENT .
Full-Day Minimum Minutes (<code>full_day_minimum_minutes</code>)	480 Minutes (8 Hours)	Total work duration required for a full day credit. Working between half-day and full-day minimums marks the day as HALF DAY .
Missing Punch Treatment (<code>missing_punch_treatment</code>)	FLAG / ABSENT / HALF DAY	Determines the system default if an employee records a check-in but forgets to check-out (or vice versa). Defaults to flagging for administrative review.
Overtime Minimum Minutes (<code>overtime_minimum_minutes</code>)	60 Minutes	Extra work duration beyond shift end required before overtime minutes start accumulating.

4.3 Biometric Hardware Ingestion: ZKTeco ADMS Push & TCP Pull

YM-CMS features a hardware-agnostic attendance ingestion architecture supporting the industry-standard ZKTeco biometric communication protocols:



4.4 Device User Mappings, Deduplication & Quarantine Replay

To protect employee biometric privacy, biometric templates (fingerprint minutiae points) are enrolled and stored exclusively on the local physical terminal. YM-CMS stores only the machine's assigned numeric user ID (`external_user_id`).

- **Effective-Dated Mappings (`AttendanceDeviceUserMapping`)**: Explicitly maps an `external_user_id` (e.g., Device ID `1042`) on a specific terminal to an active company `Employment` . Date ranges prevent historical punch misattribution if a device ID is re-assigned to another employee in the future.
- **Quarantine Queue**: If punches arrive from an unmapped device ID, the system does not drop the data. It stores the events in `attendance_raw_events` with status `QUARANTINED` . Once HR registers the mapping in the UI, clicking `Reprocess Raw Event` instantly normalizes all historical quarantined punches into the attendance records without data loss.
- **Deterministic Idempotency**: Re-uploading the same CSV file, retrying network packets, or re-running device sync produces zero duplicate punch records. Duplicate events are silently matched to existing checksum fingerprints.

4.5 Normalized CSV Import Contract & Manual Ingestion

For remote construction job sites lacking network connectivity, attendance logs exported via USB flash drive can be uploaded via the standardized CSV import interface:

Standardized Attendance CSV Specification (v1)

Header: `device_code,external_user_id,punched_at_local,timezone,direction,source_event_id`

Sample Row: `DEV-K50-01,1042,2026-08-10 08:55:00,Asia/Karachi,in,EVT-99201`

Allowed Directions: `in` , `out` , `break_out` , `break_in` (or blank).

4.6 Daily Attendance Processing & Maker-Checker Corrections

Each calendar day, the system evaluates all accepted punches against the assigned shift schedule to calculate the daily attendance state (`AttendanceRecord`):

ATTENDANCE STATE	TRIGGERING CONDITION	CALCULATION & DERIVED VALUES
PRESENT	Check-in $\leq$ Shift Start + Grace Minutes. Work Duration $\geq$ Full-Day Minimum.	Payable Day = 1.0. Late Minutes = 0. Overtime calculated if check-out exceeds shift end.
LATE	Check-in > Shift Start + Grace Minutes. Work Duration $\geq$ Full-Day Minimum.	Payable Day = 1.0. Records exact <code>late_minutes</code> for monthly late penalty deductions.

ATTENDANCE STATE	TRIGGERING CONDITION	CALCULATION & DERIVED VALUES
HALF DAY	Work Duration between Half-Day Min and Full-Day Min (or approved Half-Day leave).	Payable Day = 0.5. Half-day deduction factor applied during monthly payroll calculation.
ABSENT	No punches recorded on working day (and no approved leave), or Work Duration Half-Day Min.	Payable Day = 0.0. Marked as Unpaid Absence; deducted from gross salary during payroll.
PAID LEAVE	Approved paid leave request covering the calendar date.	Payable Day = 1.0. Consumes leave balance from Leave Ledger; zero payroll deduction.
UNPAID LEAVE	Approved unpaid leave request covering the calendar date.	Payable Day = 0.0. Creates a dedicated Unpaid Leave deduction component in payroll.
HOLIDAY / REST DAY	Date matches Company Holiday or weekly rest day on Work Calendar.	Payable Day = 1.0 (for monthly salaried staff). No punch required.

Software Navigation: Submitting & Approving an Attendance Correction

- 1 Navigate:** Open Attendance & Leave ? Attendance Corrections ? Click + New Correction.
- 2 Select Record & Propose:** Select the affected Employment and Date. Choose the proposed status (e.g., change **Absent** to **Present**) and input corrected check-in/out times and a mandatory reason (e.g., "Official site visit with client").
- 3 Submit:** Click Submit for Approval. Status becomes **PENDING**.
- 4 Maker-Checker Approval:** An independent HR Manager reviews the correction evidence and clicks Approve Correction. The underlying **AttendanceRecord** recalculates immediately while preserving the original raw biometric punches in the audit trail.

4.7 Monthly Summaries & Finalization (Immutable Payroll Input)

At the conclusion of each calendar month, HR generates an **Attendance Monthly Summary** (**AttendanceMonthlySummary**) via **BuildAttendanceMonthlySummaryAction** . This aggregates total present days, late arrivals, late minutes, half days, paid leaves, unpaid leaves, absences, and overtime hours per employee. Once reviewed, clicking **Finalize Monthly Summary** locks the record into an **immutable cryptographic snapshot** (**evidence_checksum**). The finalized summary serves as the authoritative, tamper-proof input for the Monthly Payroll calculation engine.

5. Leave Management & Double-Entry Ledger Engine (HR-3)

The Leave Management Subsystem provides a policy-driven, double-entry leave accounting framework. Instead of maintaining simple mutable integer counters on employee records, YM-CMS tracks all leave accruals, adjustments, consumptions, and cancellations through an **append-only Leave Ledger** (`LeaveLedgerEntry`).

5.1 Leave Types & Effective-Dated Policies

Leave governance is structured into two configurable tiers:

- **Leave Types** (`LeaveType`): Represent formal categories of time off (e.g., *Casual Leave*, *Annual Leave*, *Sick / Medical Leave*, *Maternity Leave*, *Unpaid Leave*). Attributes include leave unit (`day` or `hour`), paid status (`is_paid`), and encashment eligibility (`is_encashable`).
- **Leave Policies** (`LeavePolicy`): Define the mathematical rules governing a Leave Type for a specific company and effective date range:
 - **Annual Entitlement** (`annual_quota_units`): Total units allocated per year (e.g., 14 days Casual, 14 days Annual, 8 days Sick).
 - **Accrual Frequency** (`accrual_cadence`): Upfront annual allocation, monthly pro-rata accrual, or tenure-based milestone accrual.
 - **Carry-Forward Limit** (`max_carry_forward_units`): Maximum unused balance permitted to roll over into the next calendar year.
 - **Negative Balance Control** (`allow_negative_balance`): Strict boolean gate. When `false`, the system strictly blocks any leave request that exceeds the employee's current ledger balance.
 - **Attachment Mandate** (`requires_attachment`, `attachment_threshold_days`): Mandates supporting documentation (e.g., a certified medical certificate for medical leave exceeding 2 consecutive days).

5.2 Double-Entry Leave Ledger Mechanics

Every balance movement is posted as an immutable entry in `leave_ledger_entries`. The current available leave balance is derived by summing all posted debits and credits:

```
+-----+
| LEAVE LEDGER ENTRY TYPES (app/Enums/LeaveLedgerEntryType.php) |
+-----+
| 1. OPENING      (+) : Initial migration or beginning-of-year entitlement balance. |
| 2. ACCRUAL      (+) : Monthly or periodic earned leave credit.                  |
| 3. ADJUSTMENT   (+/-): Authorized manual correction by HR with recorded business justification. |
| 4. CONSUMPTION  (-) : Approved leave request deduction. Linked to leave_request_id. |
| 5. REVERSAL     (+) : Restores balance upon approved leave cancellation or early recall. |
+-----+
CURRENT AVAILABLE BALANCE = SUM(Opening + Accrual + Adjustments - Consumptions + Reversals)
```

5.3 3-Tier Leave Application Workflow

Leave processing follows a rigorous multi-tier authorization chain ensuring operational coverage and HR policy enforcement:

Software Navigation: End-to-End Leave Application & Approval Flow

- 1 **Step 1: Submission (Employee / HR Proxy):** Navigate to Attendance & Leave ? Leave Requests ? Click + New Request. Select Employment, Leave Type, Starting Date, Ending Date, and input reason. The system computes total days, checks weekend/holiday overlaps, validates available ledger balance, and attaches required medical certificates. Click Submit Request (Status: **REQUESTED**).
- 2 **Step 2: Department Manager Endorsement:** The reporting manager opens the request from their approval queue, evaluates team coverage, and clicks Manager Approve. The status transitions to **MANAGER APPROVED**.
- 3 **Step 3: HR Final Sign-Off:** HR Officer performs the final compliance check and clicks HR Approve.
- 4 **System Execution:** The system automatically: (a) Posts a **consumption** row to **leave_ledger_entries**, reducing the balance; (b) Generates or updates daily **AttendanceRecord** rows for the leave dates, setting state to **Paid Leave** or **Unpaid Leave**; (c) Marks status as **APPROVED**.

5.4 Leave Cancellation, Reversal & Downstream Payroll Impact

- **Leave Cancellation:** If an approved leave is cancelled prior to execution, HR triggers Cancel Leave Request. The system transactionally posts a **reversal** row in the Leave Ledger to restore the exact balance units and resets the daily attendance records.
- **Payroll Integration:** Approved **Paid Leave** maintains 100% payable salary. Approved **Unpaid Leave** automatically registers an unpaid day count, which is pulled into the Monthly Payroll Run as a dedicated salary deduction component.

6. Department Operations, Tasks & 6:00 PM Daily Reporting

The Department Operations Subsystem links high-level project milestones and corporate objectives to everyday staff task execution, creative/technical deliverables, multi-tier supervisory reviews, mandatory daily work check-ins, and automated employee productivity scoring.

6.1 Department Operations Architecture & Hierarchy

Department Operations is structured into an 8-level accountability hierarchy designed to eliminate communication bottlenecks and ensure daily operational transparency:

Company Tenant ? Department ? Department Teams ? Team Leads (L1 Review) ? Team Members (Assignees) ? Specialized Tasks ? Daily Reports (6 PM Cutoff) ? Productivity Analytics (0–100)

6.2 Specialized Workflow Sub-Modules

Every task in YM-CMS (`Task` , sequence `TSK-YYYY-XXXXX`) dynamically renders tailored tracking schemas based on the selected Department Team:

SPECIALIZED MODULE	TAILORED FIELDS & METADATA SCHEMA	DELIVERABLE ARTIFACTS & OUTPUT FORMATS
Social Media Marketing (<code>social_media_task_details</code>)	- Platforms: Meta, Instagram, LinkedIn, X, TikTok, YouTube. - Content Formats: Reel/Short, Static Post, Carousel, Story. 8-Step Pipeline: Idea ? Brief ? Copy ? Design ? Edit ? Client Review ? Scheduled ? Published. - Post Copy, Hashtag Banks, Target Scheduled Date/Time.	Live post URLs, Meta Business Suite scheduling links, copy documents. Direct cross-delegation to assigned Graphic Designers and Video Editors.
Graphic Design (<code>design_task_details</code>)	- Design Type: Social Graphic, Billboard, Brochure, UI Screen, Logo. - Dimensions & Aspect Ratios (1:1, 9:16, 16:9, Print A4/300DPI). - Brand Guidelines, Color Palettes, Typography Requirements. - Design Brief, Required Headlines & Body Copy.	Figma canvas links, Adobe Illustrator (.ai) / Photoshop (.psd) cloud assets, high-resolution PNG/PDF preview exports.
Video Production (<code>video_task_details</code>)	- Video Type: Promo Commercial, Site Walkthrough, Reel, Podcast. - Target Duration (seconds/minutes), Aspect Ratio (9:16 / 16:9). - Raw Footage Cloud URLs (Google Drive / Dropbox / Frame.io). - Script Timings, Voiceover Notes, B-Roll Pace, Sound FX Tone.	Cloud raw footage directories, Draft V1 / V2 review links (Frame.io/Loom), 4K/1080p Final Master Video URLs.
Web Development (<code>web_dev_task_details</code>)	- Task Type: Feature, Bug Fix, UI/UX, API, Performance, Database. - Technical Specifications & Architecture Acceptance Criteria. - QA Testing Status: Untested ? In QA ? Passed ? Failed. - Bug Count Tracking & Regression Notes.	GitHub Pull Request URLs, staging test environments, production release endpoints, automated CI/CD build logs.

SPECIALIZED MODULE	TAILORED FIELDS & METADATA SCHEMA	DELIVERABLE ARTIFACTS & OUTPUT FORMATS
Sales & Business Dev (daily_work_reports)	- Daily Outreach Metrics: Leads Received, First Contacts Made. - Outbound Call Volumes & WhatsApp Business Engagements. - Client Meetings & Construction Site Inspection Visits Booked. - Proposals Dispatched, Final Deals Closed, Total Revenue Booked.	CRM deal links, formal quotation references, signed client contract PDFs, categorized lost-lead root cause analyses.

6.3 2-Tier Quality Approval Chain

Task deliverables undergo a rigorous two-tier quality control process before being marked complete:

Software Navigation: Executing the 2-Tier Task Approval Lifecycle

1

Stage 1: Member Submission: The assignee attaches all deliverable URLs (Figma, GitHub, Drive, Loom) in the Task Attachments relation manager and clicks Submit Deliverable. Task status becomes **SUBMITTED** (100% progress).

2

Stage 2: Level-1 Team Lead Review: The Team Lead inspects the files. If quality meets team standards, the Lead clicks Review & Pass to Head (recording Lead ID and timestamp). If changes are required, the Lead clicks Request Revision, entering specific revision feedback (Status becomes **REVISION REQUIRED** and revision counter increments).

3

Stage 3: Level-2 Head Final Approval: The Department Head reviews the lead-endorsed task on the Head Operations Dashboard and clicks Approve & Complete. Status transitions to **COMPLETED** and the turnaround clock stops.

6.4 Mandatory 6:00 PM Daily Reporting Cutoff & Notifications

Every active employee is required to submit a **Daily Work Report** (DailyWorkReport) summarizing tasks worked, hours spent, deliverable links, blockers, and sales metrics before the daily cutoff:

DAILY REPORTING TIME-WINDOW:

08:00 AM

06:00:00 PM (CUTOFF)

11:59 PM

SUBMITTED ON OR BEFORE 18:00:00

SUBMITTED AFTER 18:00:00

Status: [ON TIME] (Green)

Status: [LATE] (Orange)

Full Productivity Reliability Credit

Late Tagged; Minor Score Penalty

NO SUBMISSION AT CUTOFF:

Automated Cron: php artisan work-reports:check-deadline

Status: [MISSING] (Red)

Broadcasts real-time in-app alerts to Department Head

6.5 Head Operations Dashboard & Real-Time Daily Attendance & Work Matrix

- Head Operations Dashboard** (DepartmentHeadDashboard): Real-time command center displaying total departmental staff, today's report submission rate, active tasks in progress, overdue tasks, and the 1-click Level-2 approval queue.
- Daily Attendance & Work Matrix** (DailyReportingMatrixPage): An interactive grid rendering every team member alongside their real-time daily status: **ON-TIME**, **LATE**, or **MISSING**. Department heads can filter by date and team, clicking View Report to inspect deliverables.

6.6 Composite Employee Productivity Scorecard (0–100 Mathematical Formula)

YM-CMS features an automated, objective productivity scoring algorithm implemented in CalculateEmployeeProductivityService. The score is evaluated dynamically across four performance pillars:

PRODUCTIVITY PILLAR	MAX POINTS	MATHEMATICAL EVALUATION FORMULA	OPERATIONAL PERFORMANCE OBJECTIVE
Task Completion Rate	40 Pts	$(\text{Completed Tasks} / \text{Total Assigned Tasks}) \times 40$	Rewards high output and timely delivery of assigned responsibilities.
Daily Reporting Reliability	30 Pts	$(\text{On-Time 6 PM Reports} / \text{Working Days}) \times 30$	Enforces daily reporting discipline and operational transparency.
Turnaround & Timeliness	20 Pts	$20 - (\text{Overdue Tasks} \times 5)$ [Minimum 0]	Penalizes task delays that exceed established client deadlines.
First-Time Quality Index	10 Pts	$10 - (\text{Average Revisions per Task} \times 3)$ [Min 0]	Rewards high accuracy and "First-Time Right" deliverable quality.
TOTAL SCORE	100 Pts	85-100: TOP PERFORMER ? 70-84: SOLID PERFORMER ? < 70: NEEDS IMPROVEMENT ??	

7. Employee Loans, Advances & Subledger Engine (HR-6)

Phase HR-6 establishes a formal financing subledger managing corporate lending, salary advances, amortization schedules, Treasury disbursements, and automated payroll recoveries. It completely replaces informal, untracked cash advances with strict maker-checker controls and balanced double-entry accounting.

7.1 Loans vs Advances: Architectural Distinction & Control Accounts

YM-CMS maintains a strict distinction between **Employee Loans** and **Employee Advances**:

DIMENSION	EMPLOYEE LOAN (LOAN)	EMPLOYEE ADVANCE (ADVANCE)
Business Purpose	Long-term corporate financing (e.g., vehicle loan, home assistance, medical emergency).	Short-term emergency cash advance against upcoming earned monthly salary.
Recovery Horizon	Multi-month amortization schedule (e.g., 6 to 24 monthly installments).	Immediate recovery in 1 to 3 upcoming consecutive payroll cycles.
GL Control Mapping	1113 Employee Advances / Loans Receivable (Current Asset)	1113 Employee Advances Receivable (Current Asset)
Finance Charges	Supports explicit zero-charge or approved finance fee terms.	Strictly zero interest / fee. Principal recovery only.

7.2 Amortization Schedules & Treasury Disbursement

When a financing request is approved, the system generates a versioned amortization schedule (`EmployeeFinancingInstallment`) defining installment numbers, due dates, principal amounts, finance charges, and total due amounts.

Software Navigation: Originating and Disbursing an Employee Loan / Advance

- 1 Request Origination:** Navigate to Loans & Advances ? Employee Loans & Advances ? Click + New Request. Select Employment, Financing Type (Loan vs Advance), Request Date, Principal Amount (PKR), Number of Installments, and First Due Date. Click Submit for Approval (Status: **REQUESTED**).
- 2 Finance Approval (Maker-Checker):** An authorized Finance Approver inspects the employee's tenure, existing active debt, and debt-to-income ratio, then clicks Approve Request (Status becomes **APPROVED**).
- 3 Treasury Disbursement:** Treasury opens the approved financing and clicks Disburse Funds. Select Payment Method (Company Bank Account or Cash Book), Payment Date, and Instrument Reference (Cheque / Pay Order / Online Transfer).
- 4 Automated Double-Entry Posting:** The system automatically posts a balanced Treasury Payment Voucher:

```
Dr 1113 Employee Advances Receivable (Subledger: EMP-00001) ..... PKR 60,000
Cr 1111/1112 Company Bank Account / Cash Book ..... PKR 60,000
```

Financing status transitions to **ACTIVE** and the installment schedule is activated.

7.3 Multi-Channel Recovery Engine

YM-CMS supports four distinct recovery channels for liquidating active loan and advance balances:

- Channel 1: Automated Payroll Deduction (Primary):** During monthly payroll generation, the payroll engine queries all active installments where `due_date ? payroll_period_end`. The exact installment amount is deducted from Net Salary, posting a credit against `1113 Employee Advances Receivable`.

- **Channel 2: Direct Treasury Cash/Bank Receipt:** If an employee makes an out-of-cycle direct cash repayment, Treasury records a `Treasury Receipt` allocated against the financing. The subledger applies the payment to open installments in strict chronological due-date order.
- **Channel 3: Early Lump-Sum Payoff:** An employee can settle their total remaining balance early. Treasury processes a single payoff receipt, marking all future installments as `PAID` and closing the financing as `SETTLED`.
- **Channel 4: Final Settlement Deduction:** Upon employee exit, any outstanding loan or advance principal is automatically swept into the Final Settlement statement as a mandatory recovery component.

7.4 Rescheduling, Principal Waivers & Reversals

- **Rescheduling (`RescheduleEmployeeFinancingAction`):** If an employee experiences financial hardship, HR can restructure open installments over an extended duration. Prior schedule rows are marked `SUPERSEDED` and new schedule rows are instantiated without altering historical paid records.
- **Principal Waiver (`WaiveEmployeeFinancingAction`):** Under executive board approval, unpaid principal can be formally written off. The system automatically posts a balanced General Ledger journal debiting `5090 Miscellaneous Expense (Staff Welfare)` and crediting `1113 Employee Advances Receivable`, marking the installments as `WAIVED`.
- **Reversals:** If a disbursement or repayment voucher is reversed in Treasury, the financing subledger automatically rolls back installment statuses and restores exact outstanding balance snapshots.

8. Payroll Processing, Calculation Rules & Double-Entry Accounting (HR-7)

The Payroll Subsystem combines approved employee compensation structures, finalized biometric attendance summaries, double-entry leave ledger records, active loan installment schedules, and variable performance bonuses into a deterministic monthly payroll execution and General Ledger posting engine.

8.1 Compensation History & Salary Allowance Structure

Worker remuneration is defined in `EmploymentCompensation` records. Approved compensation is effective-dated and 100% immutable. The salary structure separates earnings into specific, transparent components:

SALARY COMPONENT	CLASSIFICATION	CALCULATION BASIS & ACCOUNTING EXPENSE MAPPING
Basic Salary	Fixed Basic Earning	Contracted monthly base pay. Prorated based on payable days within the payroll month. Maps to <code>5010 Salaries Expense</code> (Admin) or <code>7000 Direct Labor</code> (Project).
House & Travel Allowance	Fixed Monthly Allowance	Standard residential and transportation allowance. Maps to <code>5011 House & Travel Allowance</code> .
Food Allowance	Fixed Monthly Allowance	Staff meal and dining subsidy. Maps to <code>5012 Food Allowance</code> .
Fuel & Vehicle Allowance	Operational Allowance	Executive vehicle fuel reimbursement. Maps to <code>5013 Fuel Allowance</code> .
Mobile & Internet Allowance	Communication Allowance	Phone and connectivity stipend for remote/site staff. Maps to <code>5014 Mobile & Internet Allowance</code> .
Site & Project Allowance	Construction Site Allowance	Special hardship/relocation allowance for active site engineers. Maps to <code>7001 Site Allowances</code> .
Other Allowances	Discretionary Allowance	Special customized stipends or utility subsidies. Maps to <code>5019 Other Staff Allowances</code> .
GROSS SALARY	Derived Total	Sum of Basic Salary + All Configured Monthly Allowances.

8.2 Effective-Dated Calculation Rules & Attendance Deductions

The monthly calculation engine evaluates deductions according to active company `PayrollCalculationRule` settings:

- Payable Basic Calculation:** Basic pay is prorated according to actual payable days:
$$Payable\ Basic = \left(\frac{Basic\ Salary}{Calendar\ Days\ in\ Month} \right) \times (Payable\ Days - Unpaid\ Absences - Unpaid\ Leaves - (0.5 \times Half\ Days))$$
- Late Arrival Penalty:** Companies can enforce minute-based or count-based late arrival deductions (e.g., 1 day salary deduction for every 3 late arrivals exceeding the grace threshold).
- Unpaid Leave & Absence Deductions:** Each unexcused absence or approved unpaid leave day deducts exactly 1 full day's basic wage rate from the employee's gross entitlement.

8.3 Variable Earning Sources: Bonus & Incentives

Variable compensation (`PayrollVariableComponent`) allows HR to award performance bonuses, eid incentives, or site completion rewards. Each variable earning requires independent submission, supporting project reference, and maker-checker approval. Approved components are pulled into the upcoming payroll run as immutable earning line items.

8.4 Monthly Payroll Run Execution & Component Traceability

Monthly payroll runs (`PayrollRun` , sequence `PR-YYYY-MM`) progress through an audited 6-stage lifecycle:

Draft (Generate Entries) ? Under Review (HR Review) ? Approved (Finance Approval) ? Posted (GL Journal Created) ? Paid (Treasury Settle) ? Locked (Immutable Archive)

Software Navigation: Generating and Finalizing a Monthly Payroll Run

- 1 **Initiate Payroll Run:** Navigate to HR Management ? Payroll Runs ? Click + New Payroll Run. Select Payroll Period (Year & Month, e.g., August 2026).
- 2 **Generate Entries:** Click Generate / Refresh Entries. The engine scans all active employments, pulls approved compensation records, checks finalized attendance summaries, pulls due loan installments, attaches approved bonuses, and compiles detailed `PayrollEntry` records with cryptographic checksums.
- 3 **Review Exceptions:** HR inspects gross-to-net calculations, attendance deductions, and bank/cash allocation splits. Any manual adjustment is recorded with mandatory justification notes.
- 4 **Submit for Review:** Click Submit for Review. Status transitions to **UNDER REVIEW**. All entry fields become read-only.
- 5 **Finance Approval (Maker-Checker):** An independent Finance Director reviews totals and clicks Approve Payroll Run.
- 6 **Post to General Ledger:** Finance clicks Post to Accounts. The engine atomically compiles and posts the official balanced Payroll Journal Voucher into the open financial period.

8.5 Double-Entry General Ledger Posting Engine & Reversal Mechanics

Posting an approved Payroll Run creates an official, balanced double-entry General Ledger journal:

```
=====
BALANCED PAYROLL GENERAL LEDGER VOUCHER (VOUCHER TYPE: PAYROLL)
=====
Dr 5010 Salaries Expense (Administrative & Management Staff) ..... PKR 450,000
Dr 5011 House & Travel Allowance Expense ..... PKR 120,000
Dr 5012 Food Allowance Expense ..... PKR 45,000
Dr 5013 Fuel & Utility Allowance Expense ..... PKR 35,000
Dr 5020 Performance Bonus & Staff Incentive Expense ..... PKR 50,000
Dr 7000 Construction Direct Labor Cost (Allocated to Project Sites) .... PKR 650,000
    Cr 2110 Salary Payable (Subledger Breakdown per Employee) ..... PKR 1,210,000
    Cr 1113 Employee Advances / Loans Receivable (Recovered Installments) ..... PKR 90,000
    Cr 2150 Income Tax Withholding Payable (Statutory Employee Tax) ..... PKR 50,000
=====
TOTAL DEBITS: PKR 1,350,000 | TOTAL CREDITS: PKR 1,350,000 | STATUS: BALANCED & POSTED
=====
```

8.6 Construction Project Cost Allocations & Treasury Settlement

- **Project Cost Allocation:** For Project Staff (`EmploymentCategory::ProjectStaff`), payroll entry creation mandates project allocation lines. Direct labor costs are charged directly to specific construction projects and cost centers, updating project budget-vs-actual cost ledgers in real time.
- **Treasury Settlement (`MarkPayrollRunPaidAction`):** When salaries are disbursed, Treasury processes payments allocated against `2110 Salary Payable`. Disbursing funds via bank transfer debits `Salary Payable` and credits `1111 Company Bank Account`, generating the official **Bank Transfer Advice Schedule**.
- **Payroll Reversal:** If an error is discovered post-posting, clicking Reverse Payroll Run automatically creates a linked reversal journal in an open period, restores loan installment schedules, and releases the run for correction and regeneration.

9. Joining Letters, Recruitment & Onboarding Foundation

The Joining Letter & Onboarding Subsystem provides a templated document generation and candidate conversion workflow. It snapshots contractual terms, validates placeholders, generates immutable offer letters, and transitions candidates into formal employees upon signed acceptance.

9.1 Joining Letter Templates & Placeholder Substitution Engine

Companies define standardized offer letter templates (`JoiningLetterTemplate`). The system features an intelligent placeholder engine that injects dynamic employee, compensation, and organizational values into the rendered document body:

PLACEHOLDER TAG	DATA SOURCE & ENTITY REFERENCE	SAMPLE RENDERED OUTPUT
<code>{{employee_name}}</code>	<code>Employee.first_name + ' ' + last_name</code>	Engr. Muhammad Hamza Khan
<code>{{designation}}</code>	<code>Employment.designation.name</code>	Senior Site Structural Engineer
<code>{{department}}</code>	<code>Employment.department.name</code>	Civil Engineering & Construction
<code>{{joining_date}}</code>	<code>Employment.joining_date</code> (Formatted)	1st September 2026
<code>{{basic_salary}}</code>	<code>EmploymentCompensation.basic_salary</code>	PKR 120,000 / month
<code>{{gross_salary}}</code>	<code>EmploymentCompensation.gross_salary</code>	PKR 185,000 / month
<code>{{work_location}}</code>	<code>Employment.workLocation.name</code>	Head Office / Bahria Town Site Store
<code>{{probation_period}}</code>	Calculated difference: start to end date	3 Months (90 Calendar Days)

Placeholder Security Validation

The rendering engine enforces strict template sanitization. Any unrecognized or malicious placeholder tag (e.g., `{{raw_eval}}`) is automatically rejected with an execution exception, preventing template injection vulnerabilities.

9.2 Draft Generation, Approval & Cryptographic Issuance

Joining Letter management enforces an audited 6-stage lifecycle:

Draft (Generate Snapshot) ? Pending Approval ? Approved (HR Director Sign-Off) ? Issued (SHA-256 Checksum Fixed) ? Accepted (Worker Onboarded)

Software Navigation: Generating, Approving & Issuing a Joining Letter

1

Generate Draft: Navigate to HR Management ? Joining Letters ? Click + New Joining Letter. Select Company Employment and Template. The system generates the encrypted body snapshot.

2

Submit: Click Submit for Approval (Status: **PENDING APPROVAL**).

3

Approve (Maker-Checker): HR Director reviews contractual clauses and clicks Approve Letter.

4

Issue: Click Issue Letter. The system generates a permanent SHA-256 hash of the letter body and locks the text against further edits (Status: **ISSUED**).

5

Record Acceptance: When the candidate signs and returns the contract, HR clicks Record Acceptance, inputting the accepting signatory's name and acceptance timestamp. The record transitions to **ACCEPTED** (100% Immutable).

10. Performance Appraisals & Disciplinary Warnings (HR-8)

Phase HR-8 delivers objective performance evaluation frameworks and structured disciplinary grievance workflows. It allows companies to track annual/quarterly KPI reviews, self-evaluations, manager ratings, and formal warning letter escalations.

10.1 KPI Master Library & 100-Point Appraisal Cycles

Performance management is structured into three integrated entities:

- **Performance KPIs** (`PerformanceKpi`): Central library of standard measurable performance indicators (e.g., *Project Delivery Timeliness*, *Code Quality & Review Rigor*, *Site Safety Compliance*, *Sales Revenue Target Achievement*).
- **Appraisal Cycles** (`AppraisalCycle`): Define formal review timeframes (e.g., *FY 2025-26 Annual Review*, *Q2 2026 Mid-Year Review*) with configured minimum and maximum rating scales (e.g., 1.00 to 5.00).
- **Performance Appraisals** (`PerformanceAppraisal`): Individual review documents. Appraisals require itemized KPI weightages totaling **exactly 100%**.

Software Navigation: Conducting an Employee Performance Appraisal

1

Initiate Appraisal: Open HR Management ? Performance Appraisals ? Click + New Appraisal. Select Active Cycle, Employee, and Lead Reviewer.

2

Allocate KPI Weights: Add appraisal items from the KPI library. Assign individual weight percentages (e.g., Technical Delivery: 40%, Safety Compliance: 30%, Team Leadership: 30%). The sum must equal 100.00%. Click Submit Appraisal.

3

Manager Evaluation & Scoring: The Reviewer grades each KPI item against the approved rating scale (1 to 5), entering detailed qualitative remarks and improvement goals. The system calculates the weighted composite score:
$$\text{Overall Score} = \sum \left(\text{Item Rating} \times \frac{\text{Item Weight}}{100} \right)$$

4

Review & Final Approval: The Reviewer clicks Submit Review. The Department Head performs the final review and clicks Approve Appraisal.

5

Employee Acknowledgement: The employee inspects their scorecard and signs off via Acknowledge Appraisal (Status: **ACKNOWLEDGED**).

10.2 Warning Letter Templates, Issuance, Response & Closure

Disciplinary actions (`EmployeeWarning`) provide a legally defensible framework for addressing workplace misconduct, chronic absenteeism, or performance failures:

WORKFLOW STEP	AUTHORIZED ACTOR	ACTION & SYSTEM BEHAVIOR
1. Issuance (Draft ? Issued)	HR Officer / Dept Head	Selects template (<code>warning_letter_templates</code>), inputs violation severity level (First Warning, Second Warning, Final Warning), incident date, detailed violation description, and remediation timeline. Clicks Issue Warning. Body is hashed and locked.
2. Employee Response	Employee / HR Proxy	Employee submits formal written explanation and defense. Attached documents are stored securely in the private document vault. Clicks Record Response.
3. Acknowledgement	Employee	Employee formally signs and acknowledges receipt of the disciplinary notice.
4. Closure / Remediation	HR Director	Evaluates employee conduct following the remediation period. Clicks Close Warning (marking satisfactory remediation or recommending further escalation). Status becomes CLOSED .

11. Employment Movements, Transfers & Promotions (HR-8)

Phase HR-8 provides a formal mechanism for executing internal employee transfers, departmental reorganizations, promotions, manager reassignments, and linked salary revisions without overwriting historical employment records.

11.1 Movement Requests (Promotion vs Transfer) & Workflow

Staff reassignments are managed through **Employment Movement Requests** (`EmploymentMovementRequest`):

- **Movement Type** (`EmploymentMovementType`):
 - **PROMOTION**: Elevation in seniority, designation level, and responsibility, typically linked to an approved salary increase.
 - **TRANSFER**: Lateral move across departments, project sites, work locations, or reporting managers.
- **Reassignment Attributes**: A movement request can modify any combination of: (1) Department, (2) Designation, (3) Reporting Manager, (4) Work Location, and (5) Employment Category.

Software Navigation: Executing an Employee Promotion / Transfer Request

- 1 **Initiate Movement Request**: Open HR Management ? Employment Movement Requests ? Click + New Request. Select Employment, Movement Type, Effective Date, and Target Reassignment attributes (New Department, New Designation, New Manager, New Work Location).
- 2 **Link Compensation Revision**: If the promotion includes a salary increment, check `Revise Compensation` and input the new Basic Salary and Allowances.
- 3 **Submit**: Click Submit for Approval (Status: **PENDING APPROVAL**).
- 4 **Executive Approval (Maker-Checker)**: Managing Director / HR Director reviews the request and clicks `Approve & Apply Movement`.
- 5 **System Execution**: The system automatically:
 - Creates an immutable before-and-after snapshot in `employment_changes`.
 - Updates the live `Employment` record with the new department, designation, manager, and location.
 - If compensation was included, creates and auto-approves a new `EmploymentCompensation` record with matching effective date, automatically closing the previous salary period.
 - Transitions movement status to **APPLIED**.

11.2 Historical Integrity & Audit Logs

YM-CMS enforces a strict non-destructive update policy. Live employment updates never overwrite historical payroll or attendance lookups. Prior payroll runs permanently retain the exact designation, department, and salary snapshots active during that historical month.

12. Fixed Asset Custody & Issuance (HR-9)

Phase HR-9 bridges corporate capital asset management (`FixedAsset`) with human resources operations. It manages the issuance, physical custody, employee acknowledgement, transfer, return inspection, damage recovery, and clearance of enterprise equipment (e.g., laptops, surveying equipment, company vehicles, mobile devices).

12.1 Custody Issuance, Single Custodian Locking & Acknowledgement

Asset issuance is tracked via **Employee Asset Custodies** (`EmployeeAssetCustody`):

- **Single Live Custodian Invariant:** An individual fixed asset can only be issued to **exactly one** active Employment at any given time. The system enforces database row locks; attempting to issue an asset currently in another worker's custody is strictly blocked.
- **Custody Lifecycle:**
Draft ? Issued (Asset Locked) ? Acknowledged (Employee Signs) ? Return Pending ? Returned (Released)

Software Navigation: Issuing a Fixed Asset to an Employee

- 1 **Create Custody Draft:** Navigate to HR Management ? Asset Issuance ? Click + New Asset Custody. Select Employment, Fixed Asset (e.g., *Dell Precision 7680 Laptop - AST-0012*), Issue Date, Expected Return Date, Physical Condition (Brand New, Excellent, Good, Fair), and list accessories (Charger, Bag, Mouse).
- 2 **Issue Asset:** Click Issue Asset. The asset's master record updates its active custodian, locks the item against other issuances, and records status as **ISSUED**.
- 3 **Employee Sign-Off:** Employee inspects the physical hardware and signs the custody form via Acknowledge Custody (Status: **ACKNOWLEDGED**).

12.2 Transfers, Return Inspections & Damage Recovery

When hardware is transferred, returned, or reported damaged, the system executes structured recovery actions:

CUSTODY EVENT	SOFTWARE ACTION & TRIGGER	SYSTEM BEHAVIOR & LEDGER UPDATE
Asset Transfer	Click Transfer Asset ? Select New Recipient Employment.	Atomically closes current custody record (<code>Returned</code>), instantiates a new custody record for the recipient (<code>Issued</code>), and updates the Fixed Asset master custodian.
Normal Return	Click Accept Return ? Verify condition & accessories.	Marks custody as RETURNED , unlocks the Fixed Asset master record, and returns equipment to the available store inventory.
Damage / Loss Reporting	Click Report Exception ? Select Type (Damage vs Loss) & input Recovery Amount.	Marks custody as EXCEPTION , logs physical condition inspection report, and forwards the Recommended Monetary Recovery to the employee's pending clearance and Final Settlement queue.

13. Separation, Clearance & Final Settlement (HR-8, HR-9, HR-10)

Phases HR-8, HR-9, and HR-10 deliver an end-to-end employee exit, departmental clearance, source-reconciled financial settlement, and General Ledger posting engine. It guarantees that no departing worker is liquidated without returning corporate assets, clearing loan obligations, settling unserved notice periods, and reconciling terminal entitlements.

13.1 Resignation vs Termination Workflows & Access Review

Employee departures are initiated via **Employment Separations** (`EmploymentSeparation`):

- **Resignation (`resignation`)**: Employee-initiated exit. Captures Resignation Date, Proposed Last Working Date, Notice Days Required vs Served, and Handover Plan. Can be withdrawn prior to executive approval.
- **Termination (`termination`)**: Company-initiated exit. Captures Termination Authority, Official Reason (Misconduct, Redundancy, Performance, Probation Non-Confirmation), Effective Date, and Severance Terms.
- **System Access Review**: Approving a separation automatically triggers a **Pending System Access Review** (`EmploymentAccessReviewStatus::Pending`), alerting IT administrators to revoke ERP credentials, VPN access, and corporate email accounts on the approved last working date.

13.2 Multi-Departmental Employee Clearance Protocol (HR-9)

Approving a separation automatically creates an **Employee Clearance** record (`EmployeeClearance`) with mandatory clearance checklist items across eight departmental areas:

CLEARANCE AREA	AUTOMATIC SYSTEM OBLIGATION CHECKS	SIGN-OFF ACTION & BLOCKING CRITERIA
1. Fixed Assets (IT / Store)	Scans <code>employee_asset_custodies</code> for active hardware.	Hard Blocker : Cannot be cleared while unreturned assets exist. Requires asset return or approved damage recovery recommendation.
2. Loans & Advances (Finance)	Queries <code>employee_financings</code> for unpaid loan principal.	Identifies exact outstanding balance. Approves deduction from final settlement payout or records cash settlement.
3. Department Handover	Verifies project documentation and knowledge transfer.	Department Head signs off on physical files, client handover, and repository access.
4. IT & Systems	Confirms deactivation of software licenses and accounts.	IT Officer confirms revocation of email, ERP, GitHub, and cloud credentials.
5. Administration & Security	Collects company ID cards, access badges, office keys.	Admin Officer confirms return of office keys, biometric unmapping, vehicle pass.
6. Leave Review (HR)	Reconciles Leave Ledger for encashable leave balance.	HR Officer confirms final leave balance units available for encashment.

13.3 Source-Reconciled Final Settlement Calculation (HR-10)

Once departmental clearance reaches **COMPLETED**, HR prepares the formal **Final Settlement** (`FinalSettlement`). The engine pulls exact, source-reconciled components:

```

=====
SOURCE-RECONCILED FINAL SETTLEMENT STATEMENT (FS-YYYY-XXXXX)
=====
EARNING COMPONENTS (+):
1. Final Earned Salary (Prorated through approved last working date) ..... PKR 65,000.00
2. Leave Encashment (12.0 Unused Annual Leave Days @ Daily Rate) ..... PKR 48,000.00
3. Notice Pay Credit (Company-waived notice period entitlement) ..... PKR 0.00
4. Gratuity / Terminal Severance Benefit (Configured statutory formula) ... PKR 150,000.00
5. Approved Unpaid Bonus / Site Completion Incentive ..... PKR 25,000.00
-----
TOTAL GROSS EARNINGS ENTITLEMENT (A) ..... PKR 288,000.00

RECOVERY & DEDUCTION COMPONENTS (-):
1. Outstanding Employee Loan Recovery (HR-6 Financing Subledger) ..... PKR 60,000.00
2. Outstanding Employee Advance Recovery (HR-6 Advance Balance) ..... PKR 15,000.00
3. Notice Shortfall Deduction (10 Unserved Notice Days @ Daily Rate) ..... PKR 40,000.00
4. Asset Damage / Loss Recovery (HR-9 Clearance Recommendation) ..... PKR 12,000.00
5. Income Tax Withholding on Terminal Benefits ..... PKR 8,000.00
-----
TOTAL TERMINAL DEDUCTIONS & RECOVERIES (B) ..... PKR 135,000.00

=====
NET SETTLEMENT PAYABLE TO EMPLOYEE (A - B) ..... PKR 153,000.00
=====

```

13.4 Final Settlement GL Posting, Reversal & Treasury Settlement

Final Settlement processing enforces maker-checker governance (Preparer \$→\$ Reviewer \$→\$ Finance Approver). Posting creates a balanced General Ledger voucher:

```

Dr 5010 Salaries Expense (Final Earned Salary) ..... PKR 65,000.00
Dr 5030 Leave Encashment Expense ..... PKR 48,000.00
Dr 5040 Gratuity / Terminal Benefits Expense ..... PKR 150,000.00
Dr 5020 Performance Bonus Expense ..... PKR 25,000.00
    Cr 1113 Employee Advances / Loans Receivable (Loan Payoff) ..... PKR 75,000.00
    Cr 5010 Salaries Expense (Notice Shortfall Recovery Credit) ..... PKR 40,000.00
    Cr 1210 Fixed Asset Clearing (Asset Damage Recovery) ..... PKR 12,000.00
    Cr 2150 Withholding Tax Payable ..... PKR 8,000.00
    Cr 2110 Salary Payable / Terminal Settlement Payable ..... PKR 153,000.00

```

Treasury liquidates the net payable via bank transfer or cheque, automatically updating status to **SETTLED** and generating the official **Full & Final Discharge Certificate**.

14. HR Reports, Analytics & Group Consolidation (HR-11)

Phase HR-11 establishes an enterprise reporting catalog, interactive HR dashboards, authorized multi-company group consolidation, and private streamed CSV/XLSX export engines.

14.1 Tenant HR Reports & Dashboard Catalog

The **HR Reports & Dashboard** page (`/admin/company/{tenant}/hr-reports`) provides instant access to comprehensive personnel analytics:

REPORT TITLE	TARGET AUDIENCE	DATA SCOPE, FILTERS & OPERATIONAL UTILITY
Employee Master List	HR Admin / Auditors	Complete personnel roster. Filters by Department, Designation, Type, Status, and Joining Date. Exports full profile metadata.
Department-Wise Roster	Executive Management	Staff breakdown and headcount distribution across organizational divisions and sub-departments.
Salary Register	Payroll / Finance	Detailed breakdown of Basic Salary, Allowances, Gross, Deductions, and Net Pay across departments and categories.
Payroll Summary	CFO / Finance Director	Executive roll-up of total monthly salary burden, cash vs bank disbursements, tax withholdings, and loan recoveries.
Project-Wise Payroll	Project Managers	Direct construction labor cost allocations broken down by active project contracts and cost centers.
Employee Loan / Advance	Finance / Treasury	Subledger statement showing original principal, disbursed dates, monthly recoveries, paid totals, and remaining balances.
Increment History	HR Director / CEO	Audit trail of salary increments, promotion dates, historical salary growth percentages, and compensation revisions.
Attendance Summary	HR / Operations	Monthly attendance matrix showing present days, late arrivals, half days, absences, paid leaves, and overtime hours.
Leave Summary	HR / Dept Heads	Leave ledger balances, annual entitlements, used days, carried-forward balances, and pending requests.
Final Settlement Report	Finance / Audit	Terminal settlement statements, clearance statuses, leave encashment, loan recoveries, and GL voucher links.

14.2 Group HR Multi-Company Consolidated Reporting

The **Group HR Reports** interface (`/admin/company/{tenant}/group-hr-reports`) allows executive directors and group HR auditors to inspect cross-company workforce metrics across all four legal companies:

- **Unique Person vs Employment Count:** Explicitly reports unique individual headcount (de-duplicated by CNIC) alongside total active employments across the group.
- **Comparative Headcount Matrix:** Side-by-side headcount, joining trends, and turnover rates across BMC Construction, YMC Construction, 7 Orbit, and 7 Orbit Medical Billing.
- **Consolidated Salary Burden:** Group-wide payroll expenditure roll-up with dedicated sensitive-data permission gates.
- **Private Streamed Exports:** All reports support instant export to CSV and Excel (XLSX) via streamed, memory-bounded download handlers with mandatory activity logging.

15. HR Data Migration, Rollback & Operational Readiness (HR-12)

Phase HR-12 delivers industrial-strength data migration pipelines, automated pre-flight integrity validation, controlled batch rollbacks, and operational readiness auditing to ensure safe production onboarding without data corruption.

15.1 Controlled CSV Migration Pipeline & Rollback Engine

Legacy HR records are imported through seven specialized migration adapters (`HrDataMigration`):

MIGRATION ADAPTER TYPE	IMPORTED DATA ENTITIES	VALIDATION & ROLLBACK CAPABILITIES
<code>departments</code>	Department Master & Parent-Child Hierarchy	Validates parent codes; prevents circular references. Rollback verifies no employments assigned.
<code>employees</code>	Employee Profiles, Employments & Compensation	Checks CNIC uniqueness, validates salary sums. Rollback deletes imported batch atomically.
<code>document_metadata</code>	Legacy Document Vault Records	Maps document types and sensitivity tiers. Preserves metadata without fabricating dummy files.
<code>leave_balances</code>	Opening Leave Ledger Balances	Posts <code>opening</code> ledger entries. Rollback deletes ledger rows if unused.
<code>financings</code>	Active Loan & Advance Opening Schedules	Establishes open installment schedules without fabricating historical GL journals.
<code>asset_custody</code>	Existing Fixed Asset Custodies	Binds assets to employments, enforcing single custodian constraints.
<code>historical_attendance</code>	Past Monthly Attendance Summaries	Imports finalized monthly summaries for historical payroll audit.

Software Navigation: Executing a 4-Stage HR Data Migration Batch

1

Upload Batch (Stage 1): Navigate to HR > HR Data Migrations > Click + New Migration. Select Migration Type and upload source CSV (Max 10MB / 10,000 rows).

2

Dry-Run Validation (Stage 2): Click Validate Batch. The engine scans every row, validates foreign keys, checks unique constraints, and outputs a detailed line-by-line error log without writing to production tables.

3

Execute Import (Stage 3): If validation passes with 0 errors, click Import Batch. Records are committed in a single atomic transaction and an immutable batch checksum is generated.

4

Controlled Rollback (Stage 4): If an issue is identified post-import, an authorized admin clicks Rollback Migration, entering a mandatory reason. The system verifies no downstream transactions depend on the data and cleanly deletes all imported rows.

15.2 HR Operational Readiness Auditing & Recovery Manifest

- HR Operational Readiness Page (`HrOperationalReadiness`):** Live pre-flight dashboard evaluating corporate readiness across 6 key pillars: (1) Organization Setup, (2) Biometric Hardware Continuity, (3) Policy & Rule Configuration, (4) Document Compliance, (5) Migration Integrity, and (6) Payroll/GL Mapping Validation.
- Cryptographic Recovery Manifest:** Scans 24 core HR database tables, computing per-row and aggregate SHA-256 hashes to instantly detect any out-of-band tampering or database corruption.

16. Roles, Authorization & Maker-Checker Security Matrix

YM-CMS enforces a comprehensive Role-Based Access Control (RBAC) architecture managed through Filament Shield and Spatie Laravel Permission. It guarantees strict segregation of duties across all workforce operations.

16.1 Pre-Configured HR Roles & Scope Separation

ROLE TITLE	OPERATIONAL DOMAIN	GRANTED CAPABILITIES & SYSTEM BOUNDARIES
Super Admin	Global System Administration	Full unrestricted access across all companies, modules, configurations, and administrative overrides.
HR Administrator	Corporate HR Management	Full management of Employee Masters, Employments, Shifts, Policies, Document Types, and HR Setup.
HR Officer	Day-to-Day HR Operations	Creates employees, uploads documents, verifies qualifications, submits leave requests, originates loans.
Attendance Administrator	Biometrics & Scheduling	Manages attendance devices, device-user mappings, raw punch imports, attendance corrections.
Payroll Officer	Compensation & Payroll Runs	Manages compensation structures, generates monthly payroll runs, exports bank advice sheets.
Department Head	Departmental Leadership	Full visibility over department staff, creates teams, delegates tasks, performs Level-2 task approvals, audits daily work matrix, conducts appraisals.
Team Lead	Supervisory Pod Lead	Monitors assigned team members, delegates tasks, performs Level-1 deliverable quality reviews, acknowledges daily reports.
Employee / Team Member	Self-Service Execution	Views assigned tasks, updates progress %, submits deliverable URLs, files mandatory 6:00 PM daily reports, submits leave applications.

16.2 Sensitive Permissions & PII Protection Boundaries

Access to confidential personnel data requires explicit, fine-grained permission capabilities:

- `ViewSensitiveData:Employee` : Controls unmasking of national CNIC numbers and personal residential details.
- `ViewMedicalInfo:Employee` : Controls access to private health records and medical certificates.
- `ViewSensitiveBankDetails:Employee` : Controls unmasking of bank account numbers and IBAN strings.
- `ViewCompensation:Employment` : Controls visibility of monthly basic salaries, allowances, and gross earnings.
- `ManageCompensation:Employment` : Authorizes creation, editing, and submission of salary structures.
- `ViewSensitiveDisciplinaryData:EmployeeWarning` : Authorizes inspection of confidential warning letters.

16.3 Segregation of Duties Matrix Across All HR Workflows

To eliminate internal fraud and administrative errors, YM-CMS strictly prohibits a single user from acting as both the initiator (Maker) and authorizer (Checker) of a transaction:

WORKFLOW ENTITY	INITIATOR (MAKER)	AUTHORIZER (CHECKER)	SEGREGATION RULE ENFORCED
Attendance Correction	Attendance Officer	HR Manager / Dept Head	Initiator cannot approve their own correction.
Leave Request	Employee / HR Proxy	Manager + HR Approver	Applicant cannot approve their own leave.

WORKFLOW ENTITY	INITIATOR (MAKER)	AUTHORIZER (CHECKER)	SEGREGATION RULE ENFORCED
Loan / Advance	HR Officer / Preparer	Finance Approver	Preparer cannot approve or disburse loan funds.
Salary Compensation	HR Officer / Preparer	HR Director / CEO	Preparer cannot approve salary structures.
Monthly Payroll Run	Payroll Preparer	Finance Approver	Preparer cannot approve or post the payroll run.
Performance Appraisal	Lead Reviewer	Department Head	Reviewer cannot provide final executive sign-off.
Employment Movement	HR Officer	Managing Director	Requester cannot execute promotions/transfers.
Final Settlement	HR Settlement Officer	Finance Approver	Settlement preparer cannot approve settlement GL post.

17. Practical Operational Scenarios & End-to-End Walkthroughs

This section provides realistic, step-by-step walkthroughs for common day-to-day HR scenarios in the YM-CMS software.

Scenario 17.1: Hiring and Onboarding a New Construction Site Engineer

Business Context: BMC Construction hires Engr. Tariq Mehmood as a *Site Structural Engineer* on a 3-month probation at a monthly gross salary of PKR 130,000, assigned to the Bahria Town Project Site.

Step-by-Step Software Execution

- 1 Create Employee Master:** Navigate to HR Management ? Employees ? Click + New Employee.
 - Personal Info: First Name = "Tariq", Last Name = "Mehmood", CNIC = "35201-9876543-1", Mobile = "0300-1234567".
 - Employment: Company = BMC Construction, Department = Civil Engineering, Designation = Site Structural Engineer, Category = Project Staff, Type = Permanent, Joining Date = 2026-08-01, Work Location = Bahria Town Site Store, Reporting To = Project Director.
 - Probation: Probation Start = 2026-08-01, Probation End = 2026-10-31, Notice Period = 30 Days.
 - Click Create. System generates code `EMP-00045`.
- 2 Set Compensation Structure:** In the Employee Profile, scroll to Compensation History ? Click + New Compensation.
 - Effective From = 2026-08-01. Basic Salary = PKR 80,000. House & Travel = PKR 25,000. Food Allowance = PKR 10,000. Site Allowance = PKR 15,000. Gross Salary = PKR 130,000.
 - Click Create ? Click Submit for Approval.
 - HR Director logs in and clicks Approve Compensation (Status: **APPROVED**).
- 3 Assign Shift & Biometrics:**
 - Navigate to Attendance & Leave ? Shift Assignments ? Assign Tariq to "Construction Site Shift" (08:00–17:00).
 - Navigate to Attendance Devices ? Device User Mappings ? Map Device User ID "1045" on Site Terminal "DEV-SITE-01" to Tariq Mehmood.
- 4 Upload Onboarding Documents:** Open Tariq's Profile ? Documents tab ? Upload scanned CNIC, Engineering Degree, and Signed Appointment Letter. HR verifies and approves each file.

Scenario 17.2: End-to-End Monthly Payroll Execution & General Ledger Posting

Business Context: It is August 31, 2026. HR processes the monthly payroll run for BMC Construction covering all 45 employees, incorporating attendance summaries, bonus incentives, and loan installment recoveries.

Step-by-Step Software Execution

- 1 **Finalize Monthly Attendance:** Navigate to Attendance & Leave ? Attendance Monthly Summaries ? Click + Generate Monthly Summary for Period August 2026. Review present days, late minutes, and unpaid absences. Click Finalize Monthly Summary (Checksum locked).
- 2 **Approve Variable Bonuses:** Open Payroll ? Bonus & Incentives. Review submitted project completion bonuses (PKR 50,000 total) and click Approve.
- 3 **Generate Payroll Run:** Open HR Management ? Payroll Runs ? Click + New Payroll Run.
 - o Period = 2026-08. Click Generate Entries.
 - o The engine automatically compiles 45 employee entries, deducting unpaid leaves, late penalties, and active loan installments (PKR 90,000 total).
 - o Review Gross Total (PKR 4,500,000) and Net Payable Total (PKR 4,250,000).
- 4 **Submit & Approve:** Click Submit for Review. Finance Director logs in, verifies the salary register, and clicks Approve Payroll Run.
- 5 **Post to General Ledger:** Finance clicks Post to Accounts. The engine atomically generates and posts the balanced Payroll Journal Voucher, crediting 2110 Salary Payable and 1113 Employee Advances Receivable.
- 6 **Treasury Payout & Payslips:** Treasury disburses funds via online bank transfer, attaches the bank reference, and marks the run as PAID. Employees receive automated PDF Salary Slips.

Scenario 17.3: Employee Resignation, Clearance, and Final Settlement

Business Context: Senior Architect Asim Riaz resigns from YMC Construction with an approved last working date of August 20, 2026. He holds an outstanding laptop custody and an active employee loan balance of PKR 40,000.

Step-by-Step Software Execution

- 1 **Record Resignation:** Navigate to HR Management ? Employment Separations ? Click + New Separation. Type = Resignation, Request Date = 2026-07-20, Last Working Date = 2026-08-20, Notice Required = 30 Days, Notice Served = 30 Days. Managing Director approves the resignation.
- 2 **Execute Departmental Clearance:** Open HR Management ? Employee Clearances ? Open Asim's Clearance Checklist:
 - o IT / Store: Asim returns the laptop in good condition. IT accepts the return, clearing the Asset obligation.
 - o Finance: Identifies PKR 40,000 outstanding loan balance. Notes loan deduction for Final Settlement.
 - o HR / Admin: Reconciles 8 days unused annual leave for encashment; collects building access card.
 - o All 8 checklist items sign off ? Clearance status becomes COMPLETED.
- 3 **Prepare Final Settlement:** Open HR Management ? Final Settlements ? Click + New Settlement ? Select Asim Riaz.
 - o Click Refresh Source Components. The engine pulls:
 - Final Earned Salary (20 Days in August) = PKR 80,000.
 - Leave Encashment (8 Days Unused Leave) = PKR 32,000.
 - Gratuity Entitlement = PKR 75,000.
 - Outstanding Loan Payoff Deduction = -PKR 40,000.
 - o Total Net Settlement Payable = PKR 147,000.
- 4 **Approve, Post & Settle:** Finance Director reviews and clicks Approve Settlement ? Click Post to Accounts (Posts balanced GL voucher and updates financing subledger to Settled). Treasury disburses PKR 147,000 via cheque, marking the settlement SETTLED.

18. Troubleshooting & Operational Quick Reference

This section provides diagnostic solutions for common operational issues and an executive quick-reference guide.

18.1 Comprehensive Troubleshooting Matrix

OPERATIONAL PROBLEM	PROBABLE ROOT CAUSE	DIAGNOSTIC PROCEDURE	RESOLUTION & VERIFICATION
Biometric punches not appearing on Attendance Records.	1. Device User ID is not mapped to Employment. 2. Shift Assignment is missing or expired.	1. Check Attendance Raw Events for QUARANTINED status. 2. Check Shift Assignments for active record.	Map user ID in Device User Mappings & click Reprocess Event. Assign active shift schedule.
Employee Code Sequence collision error.	A manual employee code was previously entered matching the next sequence counter.	Inspect Employee Code Sequences and compare against highest code in Employments.	System automatically skips collisions. If sequence is stuck, edit <code>next_number</code> to exceed highest existing code.
Leave Request blocked with "Insufficient Balance" error.	Policy has <code>allow_negative_balance = false</code> and ledger balance is 0.	Inspect Leave Ledger Entries filtered by employee and leave type.	Post an approved <code>adjustment</code> or <code>accrual</code> entry, or select Unpaid Leave.
Payroll Run fails to generate entries.	One or more active employees lack an active, approved <code>EmploymentCompensation</code> .	Check system error notification for the specific unconfigured employee code.	Create and approve a compensation record with an effective date \$le\$ payroll period start.
Asset cannot be issued to employee.	The asset is currently marked <code>Issued</code> to another custodian.	Inspect Asset Issuance records filtered by the target Fixed Asset ID.	Process a formal return or transfer from the previous custodian before issuing to the new worker.
Final Settlement cannot be prepared.	1. Separation is not approved. 2. Clearance checklist has pending mandatory items.	Check Employment Separations and Employee Clearances status.	Approve the separation and ensure all 8 departmental clearance checklist items are signed off.

18.2 System Navigation Index & URL Reference Table

FUNCTIONAL DOMAIN	FILAMENT NAVIGATION MENU PATH	DIRECT SYSTEM URL ENDPOINT
Employee Profiles	HR Management ? Employees	<code>/admin/company/{tenant}/employees</code>
Company Employments	HR Management ? Employments	<code>/admin/company/{tenant}/employments</code>
Departments & Hierarchy	HR Management ? Departments	<code>/admin/company/{tenant}/departments</code>
Designations	HR Management ? Designations	<code>/admin/company/{tenant}/designations</code>
Work Locations	HR Management ? Work Locations	<code>/admin/company/{tenant}/work-locations</code>
Compensation History	HR Management ? Compensation History	<code>/admin/company/{tenant}/employment-compensation</code>
Document Types & Compliance	HR Management ? Hr Document Types	<code>/admin/company/{tenant}/hr-document-types</code>
Biometric Devices Registry	Attendance & Leave ? Attendance Devices	<code>/admin/company/{tenant}/attendance-devices</code>

FUNCTIONAL DOMAIN	FILAMENT NAVIGATION MENU PATH	DIRECT SYSTEM URL ENDPOINT
Device User Mappings	Attendance & Leave ? Device User Mappings	/admin/company/{tenant}/attendance-device-user-mappings
Raw Attendance Events	Attendance & Leave ? Attendance Raw Events	/admin/company/{tenant}/attendance-raw-events
Daily Attendance Records	Attendance & Leave ? Attendance Records	/admin/company/{tenant}/attendance-records
Attendance Corrections	Attendance & Leave ? Attendance Corrections	/admin/company/{tenant}/attendance-corrections
Monthly Attendance Summaries	Attendance & Leave ? Attendance Monthly Summaries	/admin/company/{tenant}/attendance-monthly-summaries
Leave Requests & Approvals	Attendance & Leave ? Leave Requests	/admin/company/{tenant}/leave-requests
Leave Ledger Balances	Attendance & Leave ? Leave Ledger Entries	/admin/company/{tenant}/leave-ledger-entries
Head Operations Dashboard	Department Operations ? Head Dashboard	/admin/company/{tenant}/department-head-dashboard
Tasks & Delegations	Department Operations ? Tasks	/admin/company/{tenant}/tasks
Daily Work Reports (6 PM)	Department Operations ? Daily Work Reports	/admin/company/{tenant}/daily-work-reports
Daily Attendance & Work Matrix	Department Operations ? Daily Work Matrix	/admin/company/{tenant}/daily-reporting-matrix
Productivity Analytics	Department Operations ? Performance	/admin/company/{tenant}/employee-performance
Employee Loans & Advances	Loans & Advances ? Loans & Advances	/admin/company/{tenant}/employee-financings
Monthly Payroll Runs	HR Management ? Payroll Runs	/admin/company/{tenant}/payroll-runs
Bonus & Incentives	Payroll ? Bonus & Incentives	/admin/company/{tenant}/payroll-variable-components
Performance Appraisals	HR Management ? Performance Appraisals	/admin/company/{tenant}/performance-appraisals
Disciplinary Warnings	HR Management ? Employee Warnings	/admin/company/{tenant}/employee-warnings
Promotions & Transfers	HR Management ? Movement Requests	/admin/company/{tenant}/employment-movement-requests
Fixed Asset Custody	HR Management ? Asset Issuance	/admin/company/{tenant}/employee-asset-custodies
Separations & Exits	HR Management ? Employment Separations	/admin/company/{tenant}/employment-separations
Employee Clearances	HR Management ? Employee Clearances	/admin/company/{tenant}/employee-clearances
Final Settlements	HR Management ? Final Settlements	/admin/company/{tenant}/final-settlements
HR Reports & Analytics	Reports ? HR Reports & Dashboard	/admin/company/{tenant}/hr-reports
Payroll Reports & Registers	Reports ? Payroll & Advances	/admin/company/{tenant}/payroll-reports
Group HR Consolidation	Reports ? Group HR	/admin/company/{tenant}/group-hr-reports

FUNCTIONAL DOMAIN	FILAMENT NAVIGATION MENU PATH	DIRECT SYSTEM URL ENDPOINT
HR Data Migrations	HR ? HR Data Migrations	/admin/company/{tenant}/hr-data-migrations
HR Operational Readiness	HR ? HR Readiness	/admin/company/{tenant}/hr-operational-readiness

18.3 Key HR & Technical Terms Glossary

- **Global Employee vs Company Employment:** Global identity (`Employee`) encapsulates persistent personal PII; Company Employment (`Employment`) encapsulates company-scoped contracts, designation, payroll, attendance, and separation history.
- **ADMS Push Protocol:** Automated Data Master Server protocol used by ZKTeco biometric terminals to transmit real-time punch events via HTTP POST to the central ERP without requiring port forwarding.
- **TCP Pull Mode:** Binary socket communication on Port 4370 where the ERP connects directly to an on-premise terminal IP to download transaction logs (`ATTLOG`).
- **Double-Entry Leave Ledger:** Accounting model for time off where leave is tracked via immutable debit/credit entries (`opening` , `accrual` , `consumption` , `reversal`).
- **Maker-Checker Governance:** Security protocol requiring transactional changes (compensations, loans, corrections, payroll, settlements) to be initiated by one user and independently authorized by a separate user.
- **Single Custodian Invariant:** Rule stating that a capital Fixed Asset can only be issued to exactly one active worker at a time, locked against concurrent issuance.
- **Payable Basic Proration:** Mathematical calculation of base salary earned during a partial month based on joining date, exit date, and attendance deductions.
- **Source-Reconciled Settlement:** Final exit statement where all earning and recovery lines strictly reconcile to approved system source ledgers (Attendance, Leave, Loans, Clearance).